Software Requirements Specification

for

Smart QR-Based Access & Payment Management System

Version 1.0 approved

Prepared by:
B. Sai Shashank (1602-23-737-165)
V. Viveak (1602-23-737-191)
M. Yashwanth (1602-23-737-192)

Department of Information Technology
Vasavi College of Engineering

April 12, 2026

Revision History

Name	Date	Reason for Changes	Version
Sai Shashank B	April 2026	Initial Draft	1.0
Viveak V	April 2026	Initial Draft	1.0
Yashwanth M	April 2026	Initial Draft	1.0

Table of Contents

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the Smart QR-Based Access & Payment Management System, Version 1.0. The document serves as a reference for the development team and stakeholders, covering the complete web-based platform for contactless toll payment, parking management, and gated community access using QR code technology.

1.2 Document Conventions

This document follows the IEEE 830-1998 SRS standard. The following conventions apply:

- Bold text indicates key terms or section headings.
- Requirement identifiers follow the format REQ-XX (functional) and NFR-XX (non-functional).
- All headings are formatted in Times New Roman, 16pt Bold.
- All body text is formatted in Times New Roman, 12pt.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audiences:

- Developers & Engineers: Refer to Sections 3, 4, and 5 for technical specifications and requirements.
- Project Managers: Refer to Sections 1 and 2 for project scope and product description.
- QA Testers: Refer to Section 4, Section 5, and Appendix C for test criteria and test cases.
- System Architects: Refer to Appendix B for UML diagrams.
- Stakeholders: Refer to Sections 1.4 and 2.2 for product overview.

1.4 Product Scope

The Smart QR-Based Access & Payment Management System is a web and mobile-compatible platform designed to eliminate manual intervention at toll plazas, parking facilities, and gated communities. The platform aims to:

- Enable users to register, load a digital wallet, and receive a QR-based gate pass for contactless toll payment.
- Allow automated parking lot entry/exit with QR-based fee computation.
- Provide time-bound QR passes for gated community and township access control.
- Provide administrators and government authorities with dashboards for monitoring, rate setting, and audit.

The system does not cover physical hardware installation of QR scanners or gate barrier maintenance in Version 1.0.

1.5 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
- UML 2.5 Specification — Object Management Group (OMG).
- IBM Rational Quality Manager — Requirements and Test Case Management Tool.
- OWASP Security Standards for Web Application Development (https://owasp.org).
- Razorpay API Documentation for Payment Gateway Integration.

2. Overall Description

2.1 Product Perspective

The Smart QR-Based Access & Payment Management System is a new, self-contained web application that replaces traditional manual toll and access control processes. It operates as a standalone platform with integration points to external payment gateways and notification services. The system is composed of a web-based front-end portal, a back-end application server, a QR generation engine, and a centralized database. Users access the system via any modern web browser or mobile app, as illustrated in the Deployment Diagram in Appendix B.

2.2 Product Functions

The major functions of the system are:

- User Registration and Role-Based Authentication (User, Admin, Govt. Authority).
- QR Code Generation and Scanning for toll and parking gate access.
- Digital Wallet Management and Payment Processing (UPI/Wallet/Card).
- Toll and Parking Fee Calculation and Deduction.
- Gated Community Access Control with time-bound QR passes.
- Fraud Detection to flag duplicate or expired QR scans.
- Real-time Notifications for toll deductions, low balance, and gate events.
- Admin Dashboard for user management, toll rate configuration, and transaction reports.

2.3 User Classes and Characteristics

1. User / Vehicle Owner

Registers an account with vehicle details, loads wallet balance, and uses QR code to pass through toll gates, parking lots, or gated entries. Expected to have basic smartphone or computer literacy. Accesses the system regularly.

2. System Administrator

Manages user accounts, sets toll rates, monitors transactions, and generates system-wide reports. Has full system privileges and accesses the system daily.

3. Government / Community Authority

Approves highway toll rates, audits transactions, and manages access rules for their jurisdiction. Accesses the system periodically for audits and rate approvals.

2.4 Operating Environment

- Server: Linux-based application server with Java EE / JSP backend.
- Database: MySQL relational database for user, wallet, payment, and transaction data.
- Client: Any modern web browser (Chrome 100+, Firefox 100+, Edge 100+, Safari 15+).

- Mobile: Responsive web design supporting Android 10+ and iOS 14+.
- Network: HTTPS over the internet; stable internet connection required.
- External Services: Payment gateway (Razorpay/UPI APIs) via REST interfaces.

2.5 Design and Implementation Constraints

- The system must use HTTPS/TLS encryption for all client-server communication.
- QR codes must be time-bound and unique per transaction to prevent reuse.
- Payment processing must comply with PCI-DSS standards.
- Role-based access control (RBAC) must be enforced for all user roles.
- The front-end must follow responsive web design principles.
- Development must follow the MVC (Model-View-Controller) architectural pattern.

2.6 User Documentation

- Online Help Module: Context-sensitive help available within the portal.
- User Manual (PDF): Separate manuals for User, Admin, and Authority roles.
- Quick Start Guide: One-page reference for each user role.

2.7 Assumptions and Dependencies

- Users are assumed to have access to a stable internet connection and a smartphone or computer.
- The system depends on the availability of the third-party payment gateway API.
- QR scanner hardware at toll/parking gates is assumed to be available and functional.
- Email/SMS notification services (e.g., Firebase, Twilio) are assumed to be available.
- The hosting environment is assumed to provide a minimum 99.9% uptime SLA.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a web-based and mobile-compatible GUI. Key UI requirements include:

- A responsive login and registration page for all user roles.
- A user dashboard showing wallet balance, QR code, and transaction history.
- A payment page supporting UPI, wallet top-up, and card transactions.
- A gate pass page displaying the generated QR code for scanning.
- An admin dashboard for user management, toll rate configuration, and report generation.
- Consistent error messages, input validation, and loading indicators across all screens.

3.2 Hardware Interfaces

The system interfaces with QR scanner devices installed at toll gates, parking lots, and gated community entry points. The scanners are expected to communicate scan results to the system via API. The software does not directly control hardware but consumes scan events from the gate controller.

3.3 Software Interfaces

- Payment Gateway API (Razorpay/UPI): For processing digital payments.
- MySQL Database: For all persistent data storage.
- QR Code Library (e.g., ZXing / QRCode.js): For QR generation and decoding.
- Notification Service (Firebase / Twilio): For SMS and push notifications.

3.4 Communications Interfaces

- All communication between client and server shall use HTTPS (TLS 1.2+).
- REST APIs shall be used for all integrations with external payment and notification services.
- QR scan events from gate hardware shall be received via secure REST API calls.

4. System Features

4.1 User Registration & Authentication

4.1.1 Description and Priority

Allows vehicle owners to create accounts with their vehicle details and securely log in to the system. Priority: High.

4.1.2 Stimulus/Response Sequences

- User navigates to the registration page and enters name, email, phone, vehicle number, and password.
- System validates inputs, creates an account, and sends a confirmation notification.
- User logs in with credentials. System authenticates and redirects to the dashboard.

4.1.3 Functional Requirements

REQ-01: The system shall allow users to register with name, email, phone number, vehicle number, and password.

REQ-02: The system shall authenticate users via email and password with secure session management.

REQ-03: The system shall support role-based access for User, Admin, and Govt. Authority roles.

4.2 QR Code Generation & Scanning

4.2.1 Description and Priority

Generates a unique, time-bound QR code upon successful payment or registration, which is scanned at gate entry/exit points. Priority: High.

4.2.2 Stimulus/Response Sequences

- User completes payment. System generates a unique QR code tied to the vehicle and transaction.
- User presents QR code at the gate scanner. System validates the QR and triggers gate opening.
- System logs the entry/exit timestamp and marks the QR as used.

4.2.3 Functional Requirements

REQ-04: The system shall generate a unique, time-bound QR code for each valid transaction.

REQ-05: The system shall validate QR codes at gate entry/exit and reject expired or duplicate codes.

REQ-06: The system shall record entry and exit timestamps for every scan event.

4.3 Digital Payment Processing

4.3.1 Description and Priority

Enables users to load their wallet and pay toll/parking fees via UPI, wallet balance, or card. Priority: High.

4.3.2 Stimulus/Response Sequences

- User navigates to the wallet page and selects a top-up amount via UPI/card. Payment gateway processes and confirms.
- At toll gate, system checks wallet balance, deducts the toll fee, and displays a deduction confirmation.
- If balance is insufficient, system prompts user to top up before allowing gate access.

4.3.3 Functional Requirements

REQ-07: The system shall allow users to add money to their wallet via UPI, card, or net banking.

REQ-08: The system shall automatically deduct the applicable toll/parking fee upon successful QR validation.

REQ-09: The system shall display a transaction confirmation with amount, date, and updated balance.

4.4 Toll Gate & Parking Access Control

4.4.1 Description and Priority

Manages the complete gate access workflow for toll plazas, parking lots, and gated community entry. Priority: High.

4.4.2 Stimulus/Response Sequences

- Vehicle arrives at gate. User presents QR code. Gate system scans and calls the validation API.
- System validates tag, checks balance, deducts fee, generates gate pass, and signals gate to open.
- For gated communities, system validates time-bound QR pass and grants or denies access accordingly.

4.4.3 Functional Requirements

REQ-10: The system shall integrate with gate controllers to open/close barriers based on QR validation results.

REQ-11: The system shall compute parking fees based on entry and exit timestamps.

REQ-12: The system shall support time-bound QR passes for gated community access control.

4.5 Fraud Detection

4.5.1 Description and Priority

Monitors and flags suspicious transaction patterns and prevents reuse of expired or duplicate QR codes. Priority: Medium.

4.5.2 Stimulus/Response Sequences

- A used or expired QR code is presented at the gate. System detects duplicate and denies access.
- System detects multiple rapid transactions from the same account and flags for admin review.

4.5.3 Functional Requirements

REQ-13: The system shall reject any QR code that has already been used or has passed its expiry time.

REQ-14: The system shall flag accounts with suspicious transaction patterns for admin review.

4.6 Notifications & Alerts

4.6.1 Description and Priority

Sends real-time alerts to users for toll deductions, low wallet balance, and gate access events. Priority: Medium.

4.6.2 Stimulus/Response Sequences

- Toll is deducted. System immediately sends a payment confirmation notification to the user.
- Wallet balance drops below a threshold. System sends a low balance alert.
- Gate pass is generated. System notifies the user with QR code details.

4.6.3 Functional Requirements

REQ-15: The system shall send real-time notifications to users upon every toll deduction and QR generation.

REQ-16: The system shall alert users when wallet balance falls below a configurable threshold.

4.7 Admin Dashboard & Reports

4.7.1 Description and Priority

Provides administrators and government authorities with tools to manage users, set toll rates, and generate audit reports. Priority: Medium.

4.7.2 Stimulus/Response Sequences

- Admin logs in and views real-time transaction counts, active users, and revenue summary.
- Admin updates toll rates. System applies new rates to subsequent transactions.
- Govt. Authority logs in, reviews highway toll rates, and approves or audits transactions.

4.7.3 Functional Requirements

REQ-17: The system shall provide admins with a dashboard showing active users, transaction counts, and revenue.

REQ-18: The system shall allow admins to configure toll rates by vehicle type and route.

REQ-19: The system shall generate transaction history reports with filters for date, vehicle, and route.

REQ-20: The system shall allow government authority admins to approve highway rates and audit transaction logs.

5. Other Non-Functional Requirements

5.1 Performance Requirements

The system shall respond to all standard user actions (login, wallet top-up, QR generation) within 2 seconds under normal operating conditions. QR scan validation at gate entry shall complete within 1 second. The payment processing workflow may take up to 5 seconds to accommodate external gateway response times.

5.2 Safety Requirements

The system shall implement data backup mechanisms with a Recovery Point Objective (RPO) of 24 hours and Recovery Time Objective (RTO) of 4 hours. All financial transaction records shall be stored in a tamper-proof audit log. The system shall handle unexpected failures gracefully with user-friendly error messages.

5.3 Security Requirements

- All passwords shall be stored as salted hash values (bcrypt or equivalent).
- All sessions shall use secure, HttpOnly cookies with inactivity timeout.
- Role-based access control (RBAC) shall prevent unauthorized access to admin and authority functions.
- QR codes shall be cryptographically signed to prevent forgery.
- All payment data shall be handled exclusively by the PCI-DSS compliant payment gateway.
- The system shall be protected against OWASP Top 10 vulnerabilities including SQL injection and XSS.
- An audit log shall record all critical actions including login, payment, QR generation, and admin changes.

5.4 Software Quality Attributes

NFR ID	Attribute	Description
NFR-01	Performance	System responds to user actions within 2 seconds; QR validation within 1 second.
NFR-02	Scalability	System shall support increasing users and toll transactions without performance degradation.
NFR-03	Security	Secure authentication, cryptographically signed QR codes, encrypted payment transactions.
NFR-04	Availability	System shall maintain 99.9% uptime (24x7) with minimal scheduled maintenance windows.
NFR-05	Reliability	Accurate fee deductions, tamper-proof transaction logs, no loss of payment records.
NFR-06	Usability	Intuitive interface accessible to all user roles on both mobile and desktop.

NFR ID	Attribute	Description
NFR-07	Maintainability	Modular design allowing easy updates to toll rates, payment APIs, and access rules.

5.5 Business Rules

- Only registered and verified users with a linked vehicle number may generate a QR gate pass.
- QR codes are single-use and expire after a configurable time window.
- Toll deduction cannot proceed if wallet balance is insufficient — user must top up first.
- Admin accounts cannot be self-registered; they must be created by an existing system administrator.
- Govt. Authority accounts are provisioned only by the system admin and have read-only access to financial records.
- Parking fees are computed automatically based on entry and exit timestamps.

6. Other Requirements

Database Requirements:

- The system shall use MySQL with ACID-compliant transactions for all user, wallet, and payment data.
- All data shall be backed up daily with a minimum retention period of 90 days.
- The database shall support indexing on frequently queried fields such as vehicle number, transaction ID, and user ID.

Legal Requirements:

- The system shall provide a Privacy Policy and Terms of Service at registration.
- User data collection and processing shall comply with applicable data protection regulations (IT Act / GDPR).
- All payment-related operations shall comply with RBI guidelines for digital payments in India.

Appendix A: Glossary

The following terms and abbreviations are used throughout this document:

- SRS: Software Requirements Specification.
- QR: Quick Response — a type of matrix barcode used for encoding data.
- UML: Unified Modeling Language — standardized modeling language for software design.
- RBAC: Role-Based Access Control — security model restricting access based on user roles.
- HTTPS: Hypertext Transfer Protocol Secure — encrypted communication protocol.
- API: Application Programming Interface — protocols for integrating software applications.
- MVC: Model-View-Controller — a software architectural pattern.
- PCI-DSS: Payment Card Industry Data Security Standard.
- OWASP: Open Web Application Security Project.
- ACID: Atomicity, Consistency, Isolation, Durability — database transaction properties.
- RPO: Recovery Point Objective — maximum acceptable data loss period.
- RTO: Recovery Time Objective — maximum acceptable system recovery time.
- UPI: Unified Payments Interface — real-time payment system in India.
- JSP: JavaServer Pages — server-side technology for building web applications.

Appendix B: Analysis Models (UML Diagrams)

This appendix contains all UML diagrams developed for the Smart QR-Based Access & Payment Management System using IBM Rational Software Architect.

B.1 Use Case Diagram

The use case diagram depicts all system actors (User, Admin, Govt./Community Authority) and their interactions including account creation, QR generation, payment processing, gate access, and report generation.

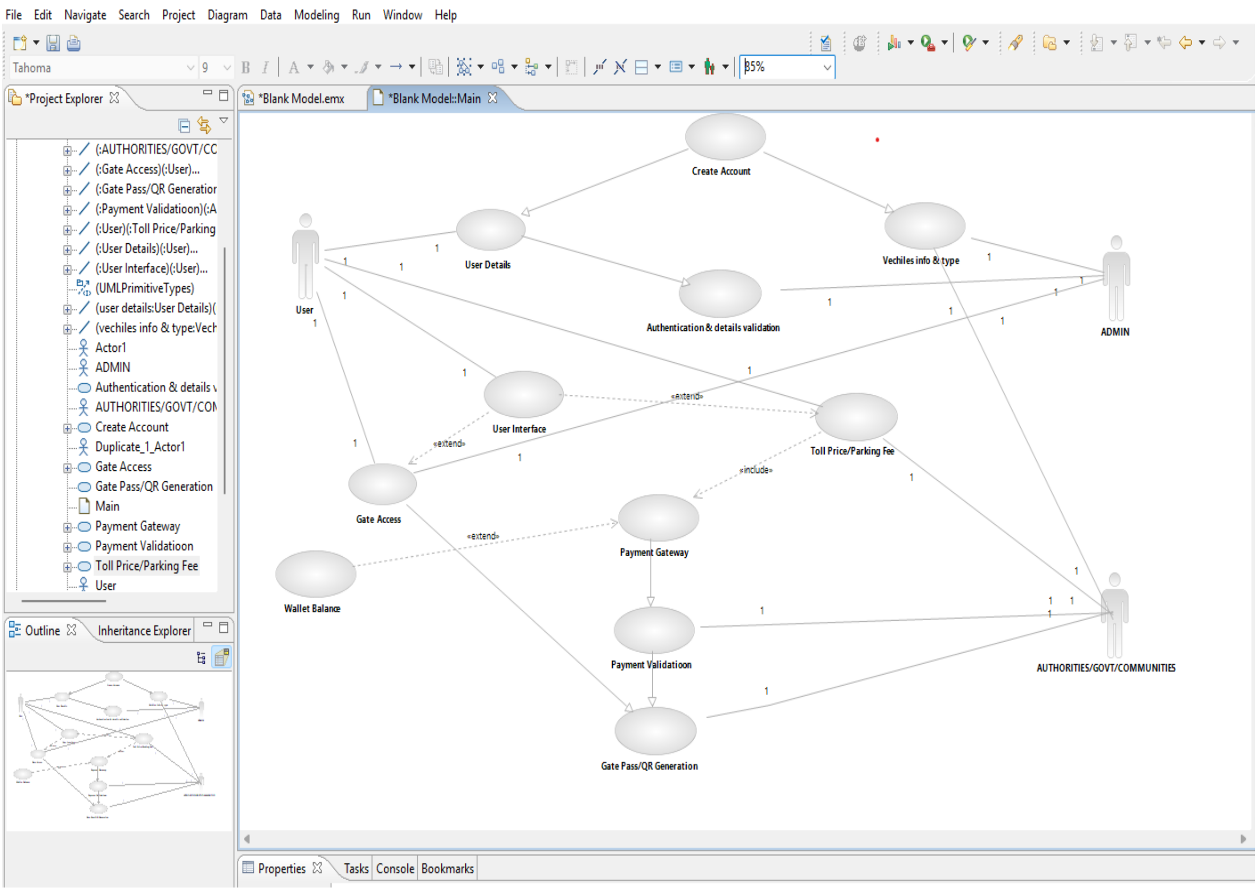

Figure B.1: Use Case Diagram — Smart QR-Based Access & Payment Management System

B.2 Class Diagram

The class diagram illustrates the static structure of the system showing key classes (User, Account, Wallet, Payment, GatePass, QR Generator, Validation, Admin, Govt. Authority Admin), their attributes, methods, and relationships.

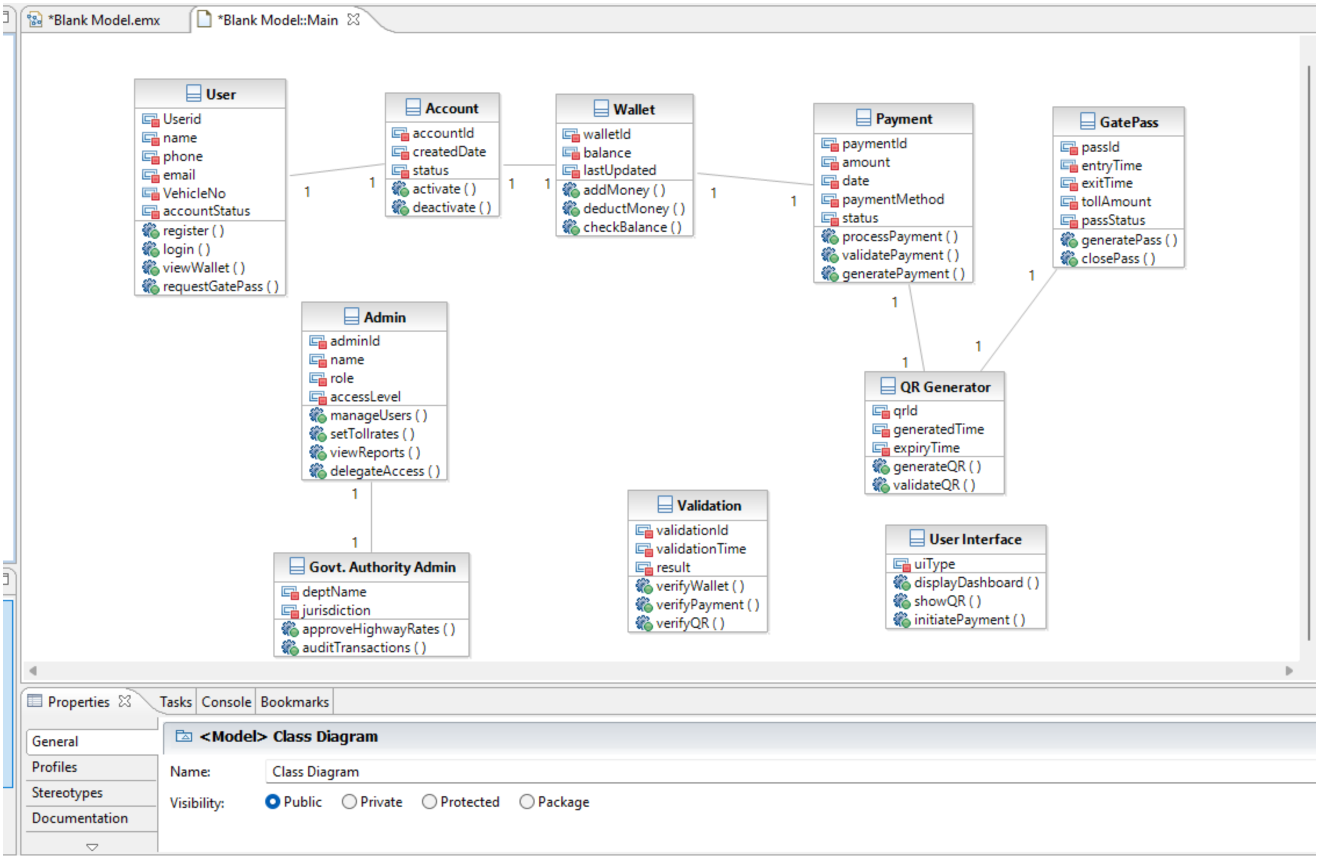

Figure B.2: Class Diagram — Smart QR-Based Access & Payment Management System

B.3 Activity Diagram (Activity / Flow Diagram)

The activity diagram illustrates the complete workflow of the system from vehicle arrival at the toll gate, QR scanning and validation, payment deduction, gate pass generation, and gate opening through to transaction logging.

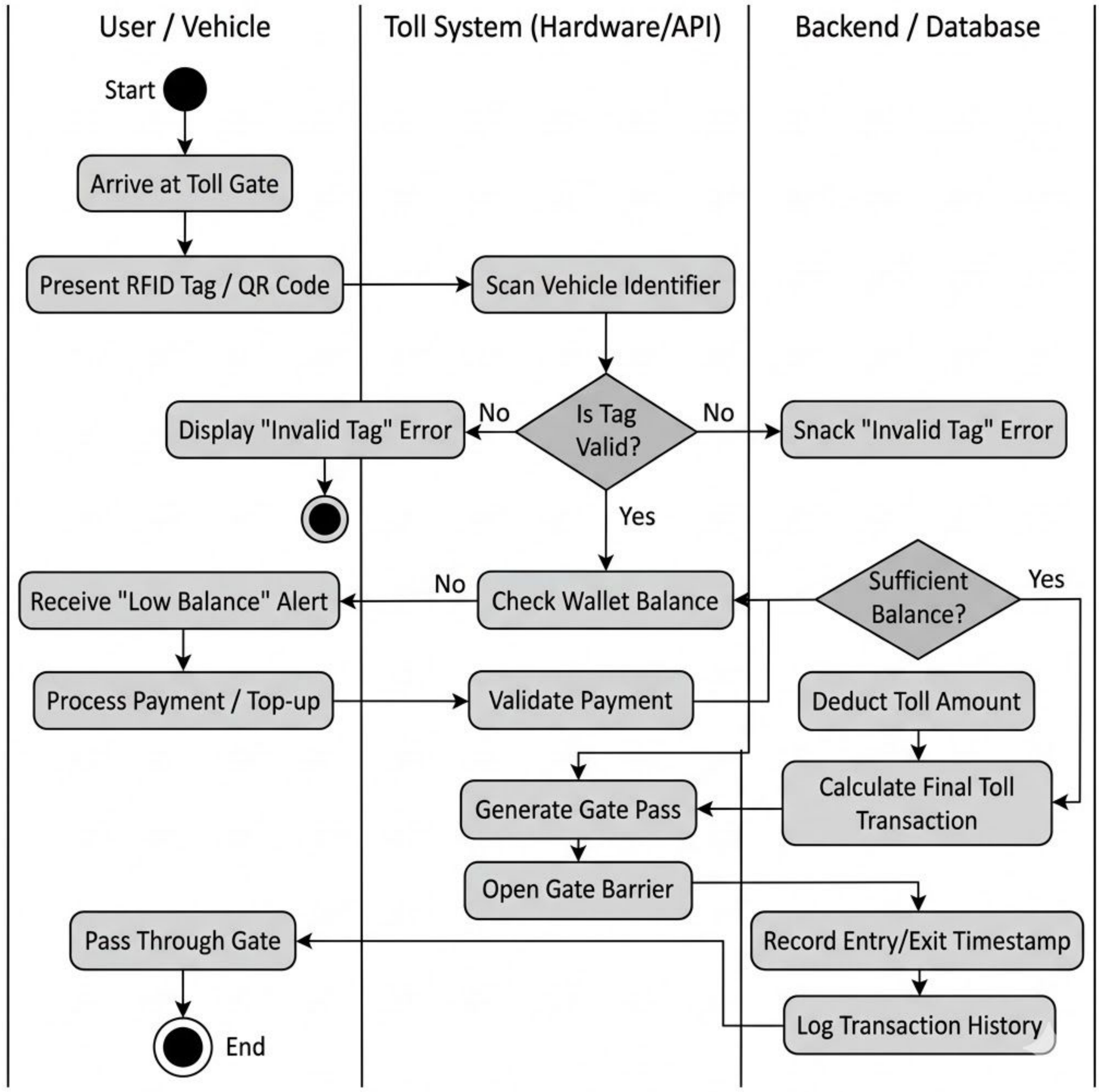

Figure B.3: Activity Diagram — Toll Gate Processing Workflow

B.4 Sequence Diagram

The sequence diagram illustrates the time-ordered interactions between actors (User, Toll Gate, Tag Validator, Wallet System, Payment System, Gate Controller, Record System) during the core toll gate transaction workflow.

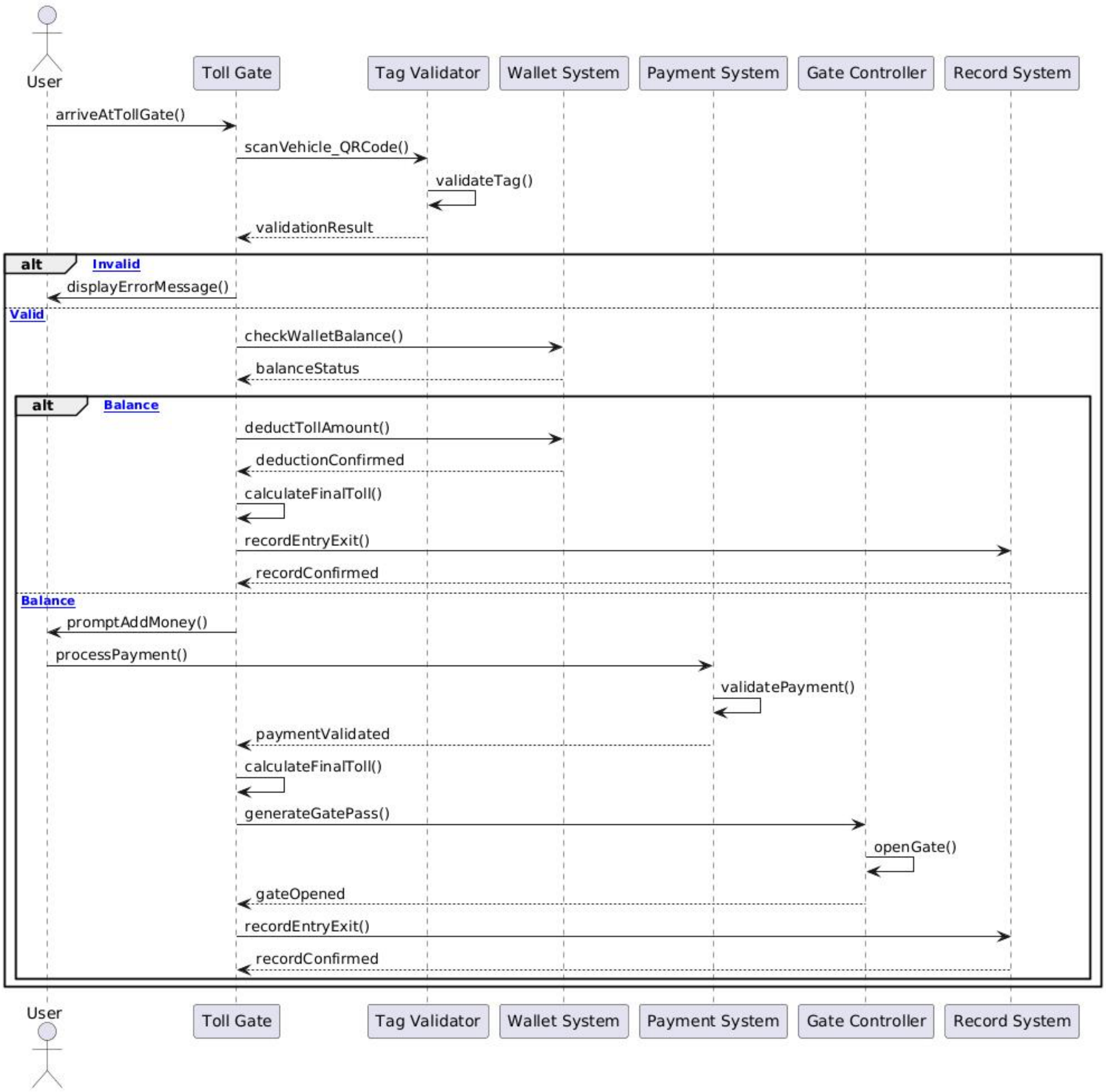

Figure B.4: Sequence Diagram — Core Transaction Workflow

B.5 Communication (Collaboration) Diagram

The communication diagram shows the structural organization of system objects and the numbered messages exchanged between them, illustrating how the User, Toll Gate, Tag Validator, Wallet System, Payment System, QR Generator, Gate Controller, and Record System collaborate.

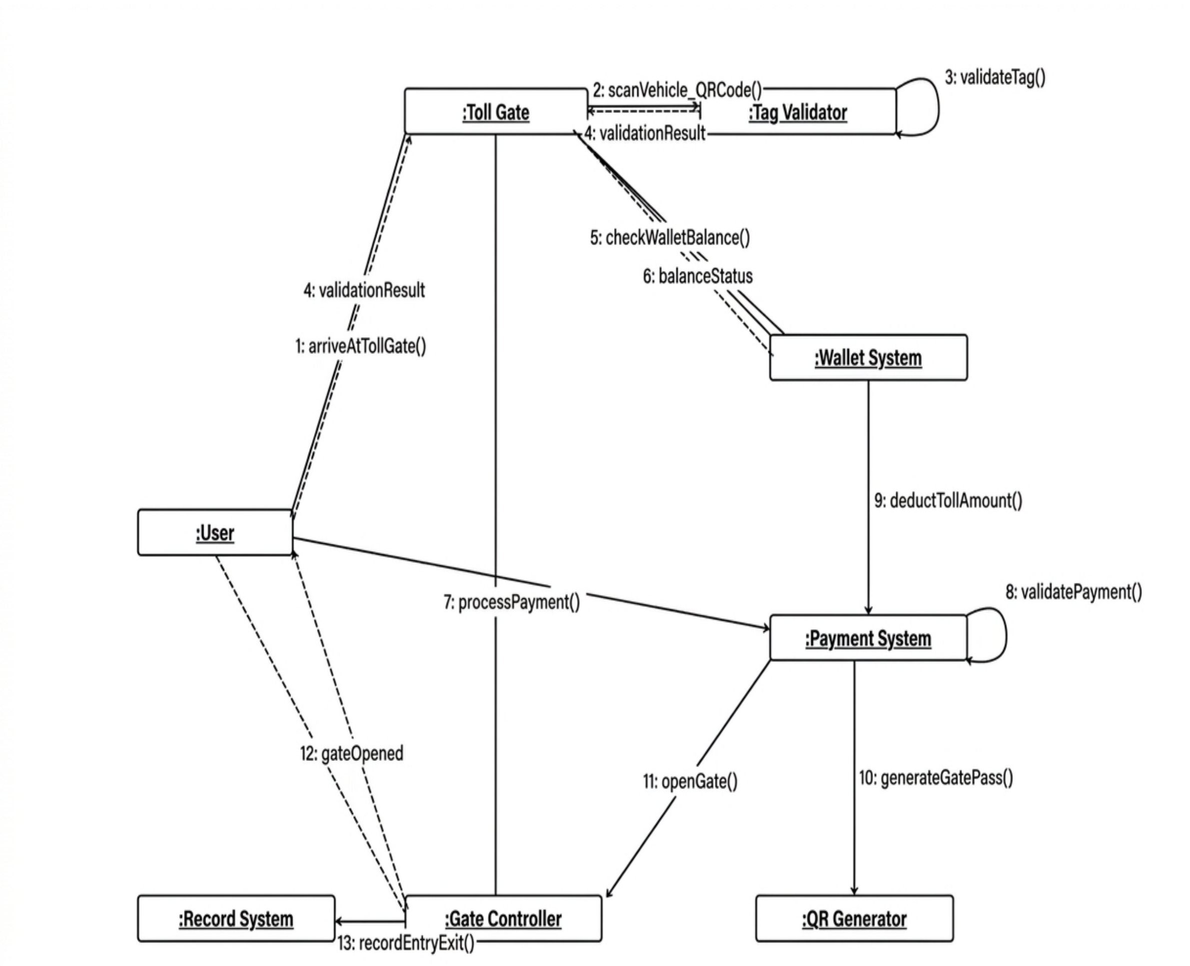

Figure B.5: Communication Diagram — Object Interactions

B.6 State Chart Diagram

The state chart diagram captures the lifecycle of a vehicle session, from Idle through QR scanning (TagScanned, Verifying, Discarded), payment processing (CheckingBalance, InsufficientBalance, SufficientBalance, DeductingToll, GatePassGenerated), gate & record management (TrafficControl, GateOperation, RecordLogging), and finally Resolved back to Idle.

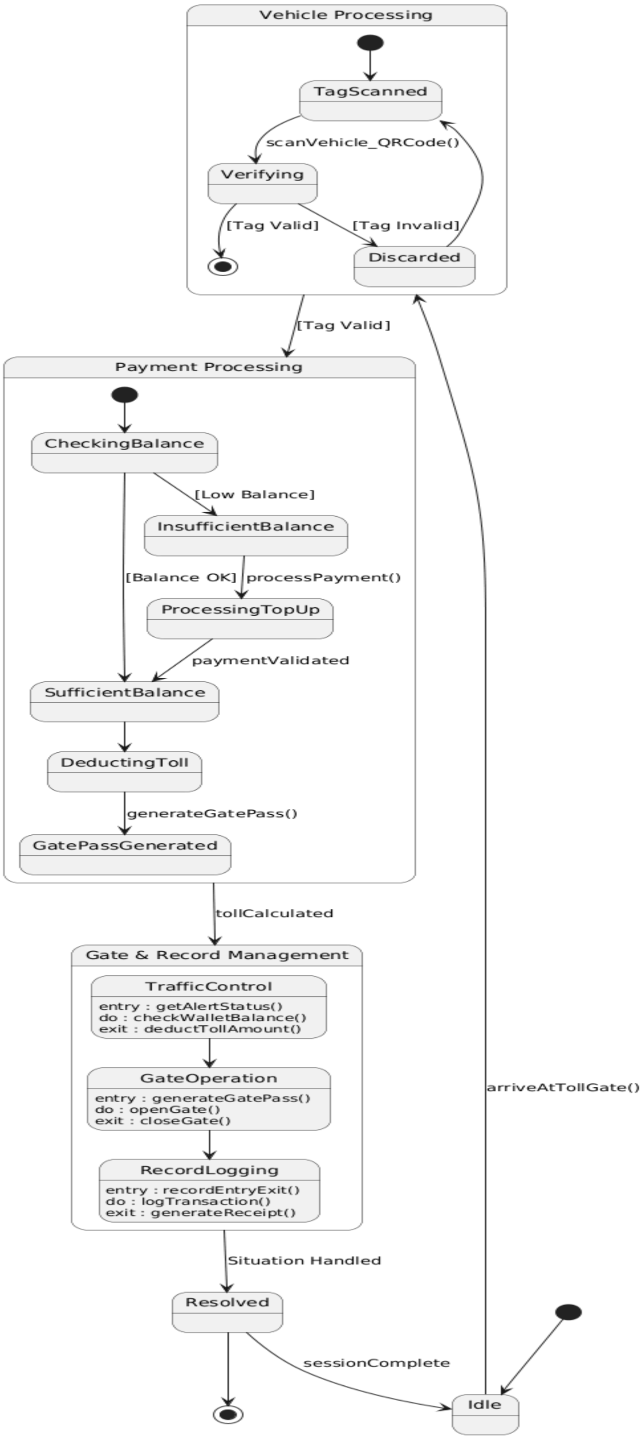

Figure B.6: State Chart Diagram — Vehicle Session Lifecycle

B.7 Component Diagram

The component diagram models the navigational components of the system: loginPage.jsp, registerPage.jsp, homePage.jsp, walletPage.jsp, tollGatePage.jsp, paymentPage.jsp, gatePassPage.jsp, transactionHistory.jsp, and adminDashboard.jsp, and their use relationships.

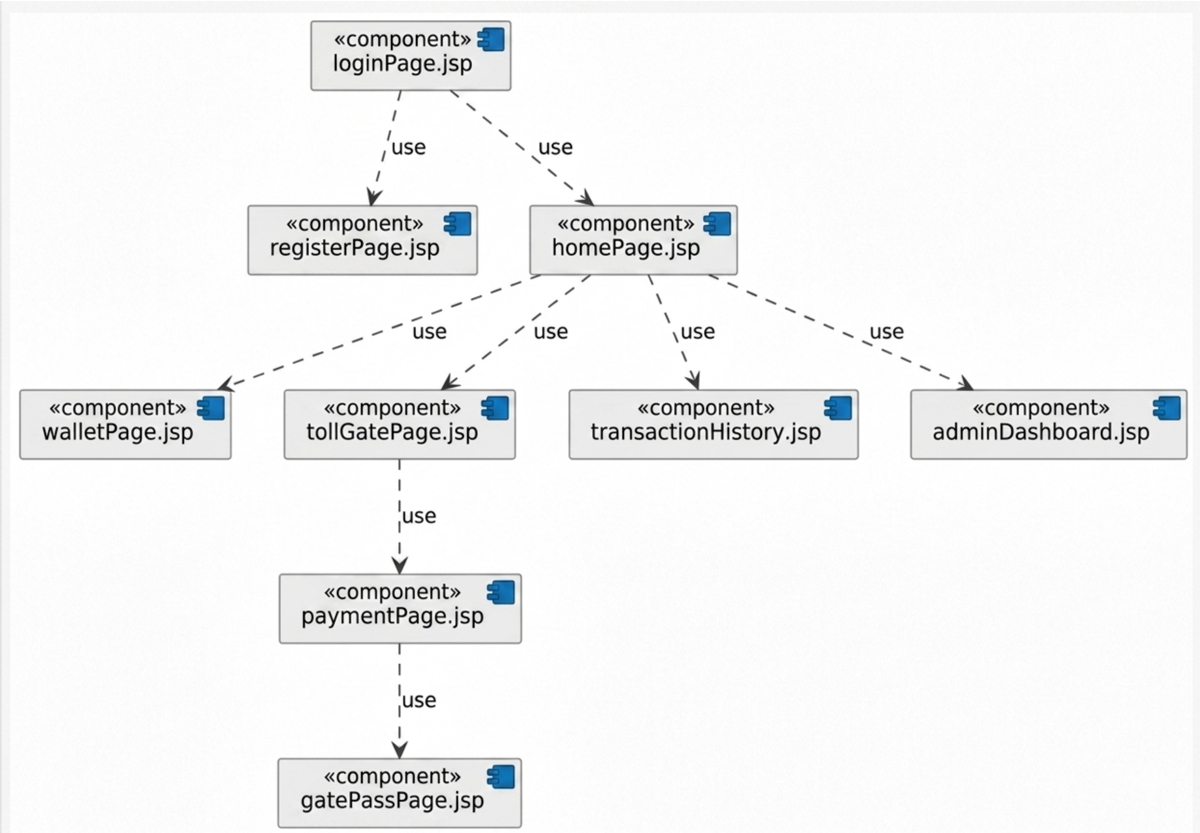

Figure B.7: Component Diagram — System Components

B.8 Deployment Diagram

The deployment diagram shows the physical architecture: client devices (Vehicle Gate Terminal, Mobile App, Admin Terminal, Reporting View) interacting with the Traffic Controller, which communicates with the Toll Gateway (JSP), Toll Logic (Services + Validation), and Database (tollDatabase + userDatabase).

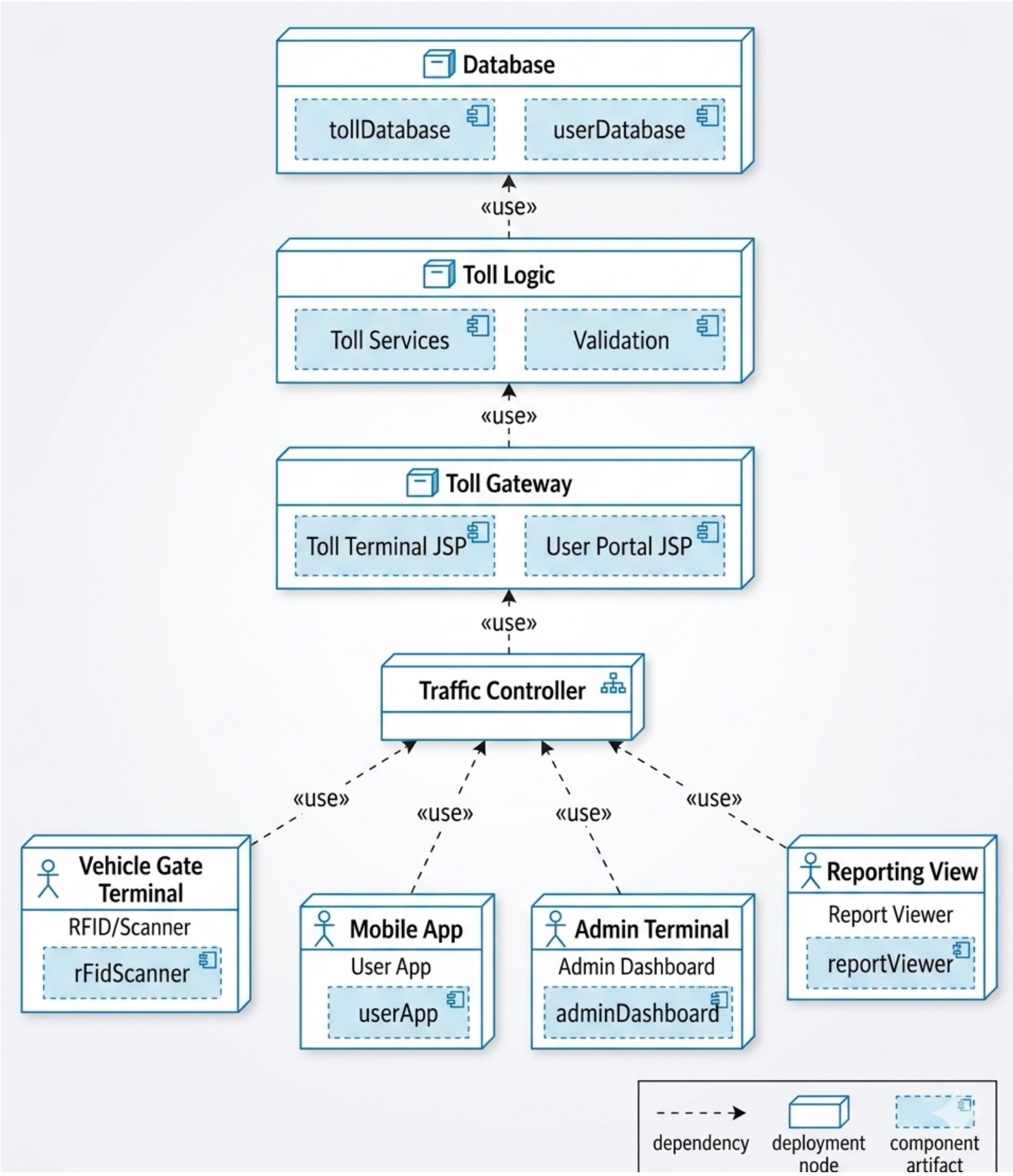

Figure B.8: Deployment Diagram — System Infrastructure

Appendix C: Test Cases

The following test cases were defined in IBM Rational Quality Manager (RQM) and mapped to the system requirements.

Test Case 1: Digital Payment via QR Code

Field	Details
Test Case ID	TC-001
Title	Digital Payment via QR Code
Originator	RQMADMIN
State	Draft
Weight	100
Input	User scans QR code and pays via UPI/card
Expected Output	Payment processed successfully. Transaction details displayed.
Linked Requirements	REQ-04: QR Code Generation (ID 110), REQ-07: Digital Payments (ID 109)

Test Case 2: High Performance & Availability

Field	Details
Test Case ID	TC-002
Title	High Performance & Availability
Originator	RQMADMIN
State	Draft
Weight	100
Input	Multiple users accessing system at peak time
Expected Output	System responds within 2-3 seconds. No downtime, system available continuously.
Linked Requirements	NFR-01: Fast Response Time (ID 117), NFR-04: 24x7 Working (ID 113)

Test Case 3: UI Usability and System Scalability

Field	Details
Test Case ID	TC-003
Title	UI Usability and System Scalability
Originator	RQMADMIN
State	Draft
Weight	100
Input	Multiple users access system on different devices (mobile/desktop). Increase number of users simultaneously.
Expected Output	Interface remains user-friendly and easy to navigate. System handles increased load without performance drop.
Linked Requirements	NFR-02: Scalable (ID 116), NFR-06: UI (ID 115)

Appendix D: To Be Determined (TBD) List

The following items remain to be determined and shall be resolved prior to final system design and development:

1. TBD-01: Final selection of payment gateway provider (Razorpay vs. PayU) and specific API version.
2. TBD-02: Specific QR code library and signing algorithm for cryptographic QR generation.
3. TBD-03: Exact notification service provider (Firebase vs. Twilio vs. AWS SNS) and delivery strategy.
4. TBD-04: Hosting environment and cloud provider (AWS / Azure / On-premise) for production deployment.
5. TBD-05: Final database version and schema migration strategy for Version 1.1 enhancements.
6. TBD-06: Detailed UI wireframes and design system (to be provided in a separate UI Specification document).
7. TBD-07: Hardware QR scanner model and gate controller API protocol specification.
8. TBD-08: Multi-jurisdiction support roadmap (multiple state toll systems) beyond Version 1.0.

--- End of Document ---

Smart QR-Based Access & Payment Management System — SRS v1.0